CHAPTER 5

LINEAR INEQUALITIES

CHAPTER ESSENCE AT A GLANCE:
Inequalities tell you which half of the universe satisfies a condition! Graph the equality line first, then pick a simple test point like (0,0) to shade the correct half-plane. The only golden rule: flip the inequality sign whenever you multiply or divide by a negative number!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Golden Sign Reversal Rule:** Multiplying or dividing an inequality by a negative number REVERSES the inequality sign! E.g. -2x ≤ 6 implies x ≥ -3.
- **Interval Notation:** x < a is (-∞, a) (open bracket); x ≥ b is [b, ∞) (closed bracket, boundary included).
- **2D Graphical Half-Planes:** Graph equality line Ax + By = C (strict <, > uses DASHED line; non-strict ≤, ≥ uses SOLID line). Test (0, 0) to shade true side.
- **Double Inequalities:** To solve a ≤ Bx + C < D, isolate x in the center by subtracting C from all 3 parts, then dividing by B.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$a < b \text{ and } c < 0 \rightarrow ac > bc \quad \text{(Sign Inversion upon Negative Scaling)}$$

Fundamental Golden Rule of Algebraic Inequalities

$$|x| \leq a \iff -a \leq x \leq a \quad (x \in [-a, a])$$

Bounded Absolute Value Symmetric Interval Criterion

$$|x| \geq a \iff x \leq -a \text{ or } x \geq a \quad (x \in (-\infty, -a] \cup [a, \infty))$$

Exterior Disjoint Domain Inequality Representation

$$ax + by \leq c \rightarrow \text{Closed Half-Plane Containing Boundary Line } ax + by = c$$

Two-Dimensional Planar Half-Space Decision Rule

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Never forget to reverse the sign when dividing by a negative number! -2x ≤ 6 becomes x ≥ -3, NOT x ≤ -3.
- **Alert:** If the boundary line passes through the origin (0,0), you cannot test with (0,0). Test with (1,0) or (0,1) instead!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Solve for real x: 4 x + 3 < 6 x + 7. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Rearranging Terms]** 4 x - 6 x < 7 - 3 ⇒ -2 x < 4. [1 Mark]

• **[Step 2: Dividing by Negative]** Dividing by -2 reverses inequality: x > 4 over (-2) ⇒ x > -2. Solution set: (-2, ∞). [1 Mark]

• **[Boxed Result: x in (-2, ∞)]**

2. Solve for real x: 3(x - 1) <= 2(x - 3). Represent the solution on a number line. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Expansion]** 3 x - 3 ≤ 2 x - 6 ⇒ 3 x - 2 x ≤ -6 + 3 ⇒ x ≤ -3. [1 Mark]

• **[Step 2: Number Line Representation]** Solution set is (-∞, -3]. On number line: draw a solid dark circle at -3 and shade heavily to the left towards -∞. [1 Mark]

• **[Boxed Result: x in (-∞, -3]]**

3. Solve the inequality for real x: x over 4 < (5x - 2) over 3 - (7x - 3) over 5. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Simplify RHS]** RHS = [5(5x - 2) - 3(7x - 3)] over 15 = [25x - 10 - 21x + 9] over 15 = (4x - 1) over 15. [1 Mark]

• **[Step 2: Cross Multiplication]** x over 4 < (4x - 1) over 15 ⇒ 15 x < 4(4x - 1) ⇒ 15 x < 16 x - 4. [1 Mark]

• **[Step 3: Solve for x]** 15 x - 16 x < -4 ⇒ -x < -4 ⇒ x > 4. Solution set: (4, ∞). [1 Mark]

• **[Boxed Result: x in (4, ∞)]**

4. Solve the system of linear inequalities and represent the solution on a number line: 3x - 7 > 2(x - 6) and 6 - x > 11 - 2x. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: First Inequality]** 3x - 7 > 2x - 12 ⇒ 3x - 2x > -12 + 7 ⇒ x > -5 ---(1). [1 Mark]

• **[Step 2: Second Inequality]** 6 - x > 11 - 2x ⇒ -x + 2x > 11 - 6 ⇒ x > 5 ---(2). [1 Mark]

• **[Step 3: Intersection of Solutions]** Intersection of x > -5 and x > 5 is x > 5. Solution set: (5, ∞). On number line: open circle at 5 and shaded to the right towards +∞. [1 Mark]

• **[Boxed Result: x in (5, ∞)]**

5. Find all pairs of consecutive odd positive integers, both of which are smaller than 10, such that their sum is more than 11. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Setup Inequalities]** Let consecutive odd positive integers be x and x + 2. Condition 1 (both smaller than 10): x + 2 < 10 ⇒ x < 8. Condition 2 (sum more than 11): x + (x + 2) > 11 ⇒ 2x + 2 > 11 ⇒ 2x > 9 ⇒ x > 4.5. [1.5 Marks]

• **[Step 2: Integer Solutions]** Since x is an odd positive integer and 4.5 < x < 8: x can be 5 or 7. For x = 5: pair is (5, 7). For x = 7: pair is (7, 9). [1.5 Marks]

• **[Boxed Result: Pairs are (5, 7) and (7, 9)]**

6. Solve the system of inequalities graphically: x + y >= 5 and x - y <= 3. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Line 1 (x + y = 5)]** Passes through (5, 0) and (0, 5). Test (0, 0): 0 ≥ 5 is FALSE. Shade the half-plane away from the origin (above or right of line). [1.5 Marks]

• **[Step 2: Line 2 (x - y = 3)]** Passes through (3, 0) and (0, -3). Test (0, 0): 0 - 0 ≤ 3 is TRUE. Shade the half-plane containing the origin (above or left of line). [1.5 Marks]

• **[Step 3: Common Region]** The solution region is the common shaded intersection of both half-planes, including boundary lines. [1 Mark]

• **[Boxed Result: Common wedge-shaped region satisfying both inequalities]**

7. Solve the system of inequalities graphically: 2x + y >= 4, x + y <= 3, and 2x - 3y <= 6. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Individual Half-Planes]** 1. 2x + y = 4: Points (2, 0), (0, 4). (0, 0) gives 0 ≥ 4 (False). Shade away from origin. 2. x + y = 3: Points (3, 0), (0, 3). (0, 0) gives 0 ≤ 3 (True). Shade towards origin. 3. 2x - 3y = 6: Points (3, 0), (0, -2). (0, 0) gives 0 ≤ 6 (True). Shade towards origin. [2.5 Marks]

• **[Step 2: Feasible Solution Region]** The solution is the bounded triangular region formed by the intersection of all three half-planes. [1.5 Marks]

• **[Boxed Result: Bounded triangular feasible region]**